

Exercise 1a (4 points): The events corresponding to the various values of $\mathbb{M}_2$ are as follows:

$$\begin{aligned}\{\mathbb{M}_2 = 2\} &: \{HHHH, HHHT, HHTH, HHTT\} , \\ \{\mathbb{M}_2 = 0\} &: \{HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT\} , \\ \{\mathbb{M}_2 = -2\} &: \{TTHH, TTHT, TTTH, TTTT\} .\end{aligned}\tag{1}$$

If we write $A_{\omega_1\omega_2}$ for the event corresponding to the first two coin tosses being ω_1 and ω_2 respectively, we then have

$$\{\mathbb{M}_2 = 2\} = A_{HH} , \quad \{\mathbb{M}_2 = 0\} = A_{HT} \cup A_{TH} , \quad \{\mathbb{M}_2 = -2\} = A_{TT} .\tag{2}$$

The σ -algebra $\sigma(\mathbb{M}_2)$ is obtained by taking all possible unions, intersections and complements of the above three sets. Similarly, we have, with $A_{\omega_1\omega_2\omega_3}$ containing sequences beginning with specific choices of $\omega_1\omega_2\omega_3$,

$$\begin{aligned}\{\mathbb{M}_3 = 3\} &: \{HHHH, HHHT\} = A_{HHH} , \\ \{\mathbb{M}_3 = 1\} &: \{HHTH, HHTT, HTHH, HTHT, THHH, THHT\} = A_{HHT} \cup A_{HTH} \cup A_{THH} , \\ \{\mathbb{M}_3 = -1\} &: \{TTHH, TTHT, THTH, THTT, HTTH, HTTT\} = A_{TTH} \cup A_{THT} \cup A_{HTT} , \\ \{\mathbb{M}_3 = -3\} &: \{TTTH, TTTT\} = A_{TTT} .\end{aligned}\tag{3}$$

Exercise 1b (6 points): $\mathcal{A}_1$ is generated by the basic sets $\{\mathbb{M}_1 = 1\}$ and $\{\mathbb{M}_1 = -1\}$:

$$\begin{aligned}\{\mathbb{M}_1 = 1\} &: \{HHH, HHT, HTH, HTT\} = A_H , \\ \{\mathbb{M}_1 = -1\} &: \{THH, THT, TTH, TTT\} = A_T .\end{aligned}\tag{4}$$

$\mathcal{A}_2$ is generated by the basic sets $\{\mathbb{M}_1 = 1\}$, $\{\mathbb{M}_1 = -1\}$, $\{\mathbb{M}_2 = 2\}$, $\{\mathbb{M}_2 = 0\}$, $\{\mathbb{M}_2 = -2\}$. We just pull the $\mathbb{M}_1$ sets from the previous equation, and give the $\mathbb{M}_2$ sets below:

$$\begin{aligned}\{\mathbb{M}_1 = 1\} &: \{HHH, HHT, HTH, HTT\} = A_H , \\ \{\mathbb{M}_1 = -1\} &: \{THH, THT, TTH, TTT\} = A_T , \\ \{\mathbb{M}_2 = 2\} &: \{HHH, HHT\} = A_{HH} , \\ \{\mathbb{M}_2 = 0\} &: \{THH, THT, HTH, HTT\} = A_{TH} \cup A_{HT} , \\ \{\mathbb{M}_2 = -2\} &: \{TTH, TTT\} = A_{TT} .\end{aligned}\tag{5}$$

Taking the intersection $\{\mathbb{M}_1 = 1\} \cap \{\mathbb{M}_2 = 0\}$, that is, $A_H \cap (A_{TH} \cup A_{HT})$ gives A_{HT} . Similarly, the intersection $\{\mathbb{M}_1 = -1\} \cap \{\mathbb{M}_2 = 0\}$ is $A_T \cap (A_{TH} \cup A_{HT}) = A_{TH}$. Thus, we have the events A_{HH} , A_{HT} , A_{TH} and A_{TT} in $\mathcal{A}_2$. In fact, these are the basic sets of the σ -algebra $\mathcal{F}_2$ of events determined by the specification of the first two coin tosses. Thus, $\mathcal{A}_2 = \mathcal{F}_2$ as σ -algebras.

Finally, $\mathcal{A}_3$ is generated by the following basic sets:

$$\begin{aligned}
\{\mathbb{M}_1 = 1\} &: \{HHH, HHT, HTH, HTT\} = A_H , \\
\{\mathbb{M}_1 = -1\} &: \{THH, THT, TTH, TTT\} = A_T , \\
\{\mathbb{M}_2 = 2\} &: \{HHH, HHT\} = A_{HH} , \\
\{\mathbb{M}_2 = 0\} &: \{THH, THT, HTH, HTT\} = A_{TH} \cup A_{HT} , \\
\{\mathbb{M}_2 = -2\} &: \{TTH, TTT\} = A_{TT} , \\
\{\mathbb{M}_3 = 3\} &: \{HHH\} = A_{HHH} , \\
\{\mathbb{M}_3 = 1\} &: \{HHT, HTH, THH\} = A_{HHT} \cup A_{HTH} \cup A_{HTT} , \\
\{\mathbb{M}_3 = -1\} &: \{TTH, THT, HTT\} = A_{TTH} \cup A_{THT} \cup A_{HTT} , \\
\{\mathbb{M}_3 = -3\} &: \{TTT\} = A_{TTT} .
\end{aligned} \tag{6}$$

Since $N = 3$, we have $A_{\omega_1\omega_2\omega_3} = \{\omega_1\omega_2\omega_3\}$. Just like in the case of $\mathcal{A}_2$, by taking appropriate intersections of the events above, we can reproduce the basic sets $A_{HHH}, A_{HHT}, A_{HTH}, A_{THH}, A_{HTT}, A_{THT}, A_{TTH}, A_{TTT}$ of $\mathcal{F}_3$, the σ -algebra determined by specifying first three coin tosses. We already have A_{HHH}, A_{TTT} . The rest are obtained as follows:

$$\begin{aligned}
A_{HHT} &= \{\mathbb{M}_2 = 2\} \cap \{\mathbb{M}_3 = 1\} , \quad A_{HTH} = \{\mathbb{M}_1 = 1\} \cap \{\mathbb{M}_2 = 0\} \cap \mathbb{M}_3 = 1\} , \\
A_{THH} &= \{\mathbb{M}_1 = -1\} \cap \{\mathbb{M}_2 = 0\} \cap \{\mathbb{M}_3 = 1\} , \quad A_{HTT} = \{\mathbb{M}_1 = 1\} \cap \{\mathbb{M}_2 = 0\} \cap \mathbb{M}_3 = -1\} , \\
A_{THT} &= \{\mathbb{M}_1 = -1\} \cap \{\mathbb{M}_2 = 0\} \cap \{\mathbb{M}_3 = -1\} , \quad A_{TTH} = \{\mathbb{M}_2 = -2\} \cap \mathbb{M}_3 = -1\} .
\end{aligned} \tag{7}$$

Thus, $\mathcal{A}_3 = \mathcal{F}_3$. Note that each of the sets above correspond to a single path of the random walk over three time steps. Thus, the σ -algebra determined by the paths up to time 3 is the same as the σ -algebra generated by a knowledge of the first 3 coin tosses.

We know that $\{\mathcal{F}_n\}$ is a filtration. Hence $\{\mathcal{A}_n\}$ is also a filtration. Or we can directly show that every set in $\mathcal{A}_i$ is in $\mathcal{A}_j$ for $j > i$.

Exercise 1c (4 points): We have to show that

$$\sum_{\underline{\omega} \in \Omega_N} \mathbb{M}_n(\underline{\omega}) \mathbb{P}\{\underline{\omega}\} = \sum_{k=-n}^n k \mathbb{P}\{\mathbb{M}_n = k\} . \tag{8}$$

A path with $\mathbb{M}_n = k$ corresponds to sequences of coin tosses with $n_H + n_T = n$ and $n_H - n_T = k$. The probability that $\mathbb{M}_n = k$ is then

$$\mathbb{P}\{\mathbb{M}_n = k\} = \binom{n}{\frac{n+k}{2}} p^{\frac{n+k}{2}} q^{\frac{n-k}{2}} . \tag{9}$$

In the sum over $\underline{\omega} \in \Omega_N$, the sum over $\omega_{n+1} \cdots \omega_N$ can be carried out independent of the value of $\mathbb{M}_n$ since these values are independent of the $(n+1)^{\text{th}}$ to N^{th} coin tosses. This sum comes out to 1, so we can restrict the actual summation to coin tosses of length n .

Let us fix the value of $\mathbb{M}_n = k$. Clearly, there are $\binom{n}{\frac{n+k}{2}}$ such terms in the summation over $\underline{\omega} \in \Omega_n$, all with the same probability $p^{\frac{n+k}{2}} q^{\frac{n-k}{2}}$. Thus, the summation can be split into sectors, each sector labelled by a single value of k . The possible values of k are $-n$ to n . This reorganizes the sum over $\underline{\omega} \in \Omega_n$ as

$$\sum_{\underline{\omega} \in \Omega_n} \mathbb{M}_n(\underline{\omega}) \mathbb{P}\{\underline{\omega}\} = \sum_{k=-n}^n k \binom{n}{\frac{n+k}{2}} p^{\frac{n+k}{2}} q^{\frac{n-k}{2}} = \sum_{k=-n}^n k \mathbb{P}\{\mathbb{M}_n = k\} . \quad (10)$$

The reorganization of the sum above only relied on the each of the paths correspond to $\mathbb{M}_n = k$ being weighted by the same value k . This can be replaced by any function $f(k)$ as well since it does not affect the reorganization. Hence, we also have

$$\mathbb{E}[f(\mathbb{M}_n)] = \sum_{k=-n}^n f(k) \mathbb{P}\{\mathbb{M}_n = k\} . \quad (11)$$

Exercise 1d (4 points): The event $\{\mathbb{M}_3 = 1\}$ consists of the sequences

$$B = \{\mathbb{M}_3 = 1\} = \{HHTH, HHTT, HTHH, HTHT, THHH, THHT\} . \quad (12)$$

The conditional expectation $\mathbb{E}[\mathbb{M}_4 \mid B] = \sum_{\underline{\omega} \in B} \mathbb{M}_4(\underline{\omega}) \mathbb{P}\{\underline{\omega} \mid B\}$ requires us to compute the conditional probabilities $\mathbb{P}\{\underline{\omega} \mid B\}$. We are given that B has happened and now B is our reduced sample space which has six elements. When each of the sequences in B have equal probability, we can compute the conditional probability directly as

$$\mathbb{P}\{\underline{\omega} \mid B\} = \begin{cases} \frac{1}{6} & \underline{\omega} \in B , \\ 0 & \underline{\omega} \notin B . \end{cases} \quad (13)$$

Otherwise, if H and T have probabilities p and q , we use the formula for conditional probability:

$$\mathbb{P}\{\underline{\omega} \mid B\} = \frac{\mathbb{P}\{\{\underline{\omega}\} \cap B\}}{\mathbb{P}\{B\}} . \quad (14)$$

We have $\mathbb{P}\{B\} = ppqp + ppqq + pqpp + qpqq + qppp + qppq = 3p^2q$ and Hence, we have

$\underline{\omega}$	HHTH	HHTT	HTHH	HTHT	THHH	THHT
$\mathbb{P}\{\underline{\omega} \cap B\}$	$ppqp$	$ppqq$	$pqpp$	$pqpq$	$qppp$	$qppq$
$\mathbb{P}\{\underline{\omega} \cap B\} / \mathbb{P}\{B\}$	$\frac{p}{3}$	$\frac{q}{3}$	$\frac{p}{3}$	$\frac{q}{3}$	$\frac{p}{3}$	$\frac{q}{3}$

$$\mathbb{E}[\mathbb{M}_4 \mid B] = \sum_{\underline{\omega} \in B} \mathbb{M}_4(\underline{\omega}) \mathbb{P}\{\underline{\omega} \mid B\} = 2\frac{p}{3} + 0\frac{q}{3} + 2\frac{p}{3} + 0\frac{q}{3} + 2\frac{p}{3} + 0\frac{q}{3} = 2p . \quad (15)$$

To use the formula $\mathbb{E}[\mathbb{Y} \mid B] = \sum_{y \in \text{Ran}_B(\mathbb{Y})} y \mathbb{P}\{\mathbb{Y} = y \mid B\}$, we need $\text{Ran}_B(\mathbb{M}_4)$ which is the set of values $\mathbb{M}_4$ can take when restricted to sequences in B :

$$\text{Ran}_B(\mathbb{M}_4) = \{2, 0\} , \quad (16)$$

and the conditional probabilities $\mathbb{P}\{\mathbb{M}_4 \mid B\}$:

$$\begin{aligned}\mathbb{P}\{\mathbb{M}_4 = 0 \mid \mathbb{M}_3 = 1\} &= \frac{\mathbb{P}\{\mathbb{M}_4 = 0, \mathbb{M}_3 = 1\}}{\mathbb{P}\{\mathbb{M}_3 = 1\}} = \frac{3p^2q^2}{3p^2q} = q , \\ \mathbb{P}\{\mathbb{M}_4 = 2 \mid \mathbb{M}_3 = 1\} &= \frac{\mathbb{P}\{\mathbb{M}_4 = 2, \mathbb{M}_3 = 1\}}{\mathbb{P}\{\mathbb{M}_3 = 1\}} = \frac{3p^3q}{3p^2q} = p .\end{aligned}\quad (17)$$

Thus, we have

$$\mathbb{E}[\mathbb{M}_4 \mid B] = \sum_{y \in \text{Ran}_B(\mathbb{M}_4)} y \mathbb{P}\{\mathbb{M}_4 = y \mid B\} = 2p + 0q = 2p , \quad (18)$$

which agrees with the value obtained by the other formula for conditional expectation.

Exercise 1e (6 points): The event B^c is $\{\mathbb{M}_3 = 3\} \cup \{\mathbb{M}_3 = -1\} \cup \{\mathbb{M}_3 = -3\}$ and contains the sequences:

$$\begin{aligned}HHHH , HHHT , TTHH , TTHT , THTH , \\ THTT , HTTH , HTTT , TTTH , TTTT .\end{aligned}\quad (19)$$

$\mathbb{P}\{B^c\} = p^4 + p^3q + q^2p^2 + q^2pq + qpqp + qpq^2 + pq^2p + pq^3 + q^3p + q^4 = p^3 + 3q^2p + q^3$. This can also be written as $1 - 3p^2q = 1 - \mathbb{P}\{\mathbb{M}_3 = 1\}$.

We have $\text{Ran}_{B^c}(\mathbb{M}_4) = \{4, 2, 0, -2, -4\}$. The conditional probabilities are

$$\begin{aligned}\mathbb{P}\{\mathbb{M}_4 = 4 \mid B^c\} &= \frac{\mathbb{P}\{\{\mathbb{M}_4 = 4\} \cap B^c\}}{\mathbb{P}\{B^c\}} = \frac{p^4}{1 - 3p^2q} , \\ \mathbb{P}\{\mathbb{M}_4 = 2 \mid B^c\} &= \frac{\mathbb{P}\{\{\mathbb{M}_4 = 2\} \cap B^c\}}{\mathbb{P}\{B^c\}} = \frac{p^3q}{1 - 3p^2q} , \\ \mathbb{P}\{\mathbb{M}_4 = 0 \mid B^c\} &= \frac{\mathbb{P}\{\{\mathbb{M}_4 = 0\} \cap B^c\}}{\mathbb{P}\{B^c\}} = \frac{p^2q^2 + qpqp + pqqp}{1 - 3p^2q} = \frac{3p^2q^2}{1 - 3p^2q} , \\ \mathbb{P}\{\mathbb{M}_4 = -2 \mid B^c\} &= \frac{\mathbb{P}\{\{\mathbb{M}_4 = -2\} \cap B^c\}}{\mathbb{P}\{B^c\}} = \frac{q^2pq + qpq^2 + pq^3 + q^3p}{1 - 3p^2q} = \frac{4pq^3}{1 - 3p^2q} , \\ \mathbb{P}\{\mathbb{M}_4 = -4 \mid B^c\} &= \frac{\mathbb{P}\{\{\mathbb{M}_4 = -4\} \cap B^c\}}{\mathbb{P}\{B^c\}} = \frac{q^4}{1 - 3p^2q} .\end{aligned}\quad (20)$$

We thus have

$$\begin{aligned}\mathbb{E}[\mathbb{M}_4 \mid B^c] &= \sum_{y \in \text{Ran}_{B^c}(\mathbb{M}_4)} y \mathbb{P}\{\mathbb{M}_4 = y \mid B^c\} , \\ &= 4 \frac{p^4}{1 - 3p^2q} + 2 \frac{p^3q}{1 - 3p^2q} + 0 \frac{3p^2q^2}{1 - 3p^2q} + (-2) \frac{4pq^3}{1 - 3p^2q} + (-4) \frac{q^4}{1 - 3p^2q} .\end{aligned}\quad (21)$$

Similarly, $\mathbb{E}[\mathbb{M}_4 \mid \emptyset] = 0$ since we know that no sequence has occurred, and

$$\begin{aligned}
\mathbb{E}[\mathbb{M}_4 \mid \Omega] &= \mathbb{E}[\mathbb{M}_4] = 4p^4 + 2\binom{4}{1}p^3q + 0\binom{4}{2}p^2q^2 + (-2)\binom{4}{3}pq^3 + (-4)\binom{4}{4}q^4, \\
&= 4(p^4 - q^4) + 2\binom{4}{1}(p^3q - pq^3), \\
&= 4(p^2 - q^2)(p^2 + q^2) + 8pq(p^2 - q^2), \\
&= 4(p - q).
\end{aligned} \tag{22}$$

Exercise 1f (4 points): The events in the σ -algebra $\sigma(\mathbb{M}_2)$ are the unions, intersections and complements of the basic sets corresponding to $\{\mathbb{M}_2 = 2\}$, $\{\mathbb{M}_2 = 0\}$ and $\{\mathbb{M}_2 = -2\}$ respectively:

$$A_{HH} = \{HHH, HHT\}, \quad A_{TH} \cup A_{HT} = \{THH, THT, HTH, HTT\}, \quad A_{TT} = \{TTH, TTT\}. \tag{23}$$

These constitute a partition of the sample space Ω_3 since their union gives Ω_3 and the sets are mutually disjoint. The σ -algebra is then

$$\{A_{HH}, A_{HT} \cup A_{TH}, A_{TT}, A_{HH} \cup A_{TT}, A_{HH} \cup A_{HT} \cup A_{TH}, A_{TT} \cup A_{HT} \cup A_{TH}, \emptyset, \Omega_3\}. \tag{24}$$

It is easy to see that it is not possible to get a more refined partition of Ω_3 using the events in $\sigma(\mathbb{M}_2)$ because the other events in $\sigma(\mathbb{M}_2)$ (except $\emptyset$) can be described purely as unions of the basic sets given above.

The conditional expectation for the basic sets are computed using the method discussed above. Let us label the basic sets as

$$B_1 = A_{HH}, \quad B_2 = A_{TH} \cup A_{HT}, \quad B_3 = A_{TT}. \tag{25}$$

Let us use the formula for conditional expectations which sums over the range of the random variable restricted to one of the basic events. We need

$$\text{Ran}_{B_1}(\mathbb{M}_3) = \{3, 1\}, \quad \text{Ran}_{B_2}(\mathbb{M}_3) = \{1, -1\}, \quad \text{Ran}_{B_3} = \{-1, -3\}. \tag{26}$$

Next, we need

$$\mathbb{P}\{B_1\} = p^3 + p^2q = p^2, \quad \mathbb{P}\{B_2\} = qp^2 + qpq + pqp + pq^2 = 2pq, \quad \mathbb{P}\{B_3\} = q^2p + q^3 = q^2. \tag{27}$$

The conditional probabilities are

$$\begin{aligned}
\mathbb{P}\{\mathbb{M}_3 = 3 \mid B_1\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = 3\} \cap B_1\}}{\mathbb{P}\{B_1\}} = \frac{p^3}{p^2} = p , \\
\mathbb{P}\{\mathbb{M}_3 = 1 \mid B_1\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = 1\} \cap B_1\}}{\mathbb{P}\{B_1\}} = \frac{p^2 q}{p^2} = q , \\
\mathbb{P}\{\mathbb{M}_3 = 1 \mid B_2\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = 1\} \cap B_2\}}{\mathbb{P}\{B_2\}} = \frac{2p^2 q}{2pq} = p , \\
\mathbb{P}\{\mathbb{M}_3 = -1 \mid B_2\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = -1\} \cap B_2\}}{\mathbb{P}\{B_2\}} = \frac{2pq^2}{2pq} = q , \\
\mathbb{P}\{\mathbb{M}_3 = -1 \mid B_3\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = -1\} \cap B_3\}}{\mathbb{P}\{B_3\}} = \frac{pq^2}{q^2} = p , \\
\mathbb{P}\{\mathbb{M}_3 = -3 \mid B_3\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = -1\} \cap B_3\}}{\mathbb{P}\{B_3\}} = \frac{q^3}{q^2} = q .
\end{aligned} \tag{28}$$

Note that some of the above answers can be guessed without too much thought. For instance, there is only one possibility $\mathbb{M}_3 = -3$ can be achieved starting from $\mathbb{M}_2 = -2$, which has probability q .

We then get

$$\begin{aligned}
\mathbb{E}[\mathbb{M}_3 \mid B_1] &= \sum_{y \in \text{Ran}_{B_1}(\mathbb{M}_3)} y \mathbb{P}\{\mathbb{M}_3 = y \mid B_1\} = 3p + 1q = 2 + p - q , \\
\mathbb{E}[\mathbb{M}_3 \mid B_2] &= \sum_{y \in \text{Ran}_{B_2}(\mathbb{M}_3)} y \mathbb{P}\{\mathbb{M}_3 = y \mid B_2\} = 1p + (-1)q = 0 + p - q , \\
\mathbb{E}[\mathbb{M}_3 \mid B_3] &= \sum_{y \in \text{Ran}_{B_3}(\mathbb{M}_3)} y \mathbb{P}\{\mathbb{M}_3 = y \mid B_3\} = (-1)p + (-3)q = -p - 3q = -2 + p - q .
\end{aligned} \tag{29}$$

This means that the random variable $\mathbb{E}[\mathbb{M}_3 \mid \sigma(\mathbb{M}_2)]$ has the following values on the sequences in sample space Ω_3 :

$$\mathbb{E}[\mathbb{M}_3 \mid \sigma(\mathbb{M}_2)](\underline{\omega}) = \begin{cases} 2 + p - q & \underline{\omega} \in \{\mathbb{M}_2 = 2\} \\ p - q & \underline{\omega} \in \{\mathbb{M}_2 = 0\} \\ -2 + p - q & \underline{\omega} \in \{\mathbb{M}_2 = -2\} \end{cases} \tag{30}$$

Clearly, the random variable is $\mathbb{M}_2(\underline{\omega}) + p - q$, a result which we obtained by a different way of calculating the expectation value in class. When the random walk is symmetric, the conditional expectation of $\mathbb{M}_3$ with respect to $\sigma(\mathbb{M}_2)$ is simply $\mathbb{M}_2$. This is expected since the symmetric random walk is a martingale.

Exercise 1g (4 points): Recall that we showed in Exercise 1b that $\mathcal{A}_2$ is the same as $\mathcal{F}_2$, the σ -algebra determined by the first two coin tosses. We also take the total number of

coin tosses to be $N = 3$. The most-refined partition of $\mathcal{A}_2$ contains the basic events

$$\begin{aligned} A_{HH} &= \{HHH, HHT\} , & A_{HT} &= \{HTH, HTT\} , \\ A_{TH} &= \{THH, THT\} , & A_{TT} &= \{TTH, TTT\} , \end{aligned} \quad (31)$$

since these are the basic events which generate the rest of $\mathcal{F}_2$ which is the same as $\mathcal{A}_2$.

We run through the standard machinery to compute $\mathbb{E}[\mathbb{M}_3 \mid \mathcal{A}_2]$. First, we need

$$\begin{aligned} \text{Ran}_{A_{HH}}(\mathbb{M}_3) &= \{3, 1\} , & \text{Ran}_{A_{HT}}(\mathbb{M}_3) &= \{1, -1\} , \\ \text{Ran}_{A_{TH}}(\mathbb{M}_3) &= \{1, -1\} , & \text{Ran}_{A_{TT}} &= \{-1, -3\} . \end{aligned} \quad (32)$$

Next, we need

$$\begin{aligned} \mathbb{P}\{A_{HH}\} &= p^3 + p^2q = p^2 , & \mathbb{P}\{A_{HT}\} &= pqp + pq^2 = pq , \\ \mathbb{P}\{A_{TH}\} &= qpq + qp^2 = pq , & \mathbb{P}\{A_{TT}\} &= q^2p + q^3 = q^2 . \end{aligned} \quad (33)$$

The conditional probabilities are

$$\begin{aligned} \mathbb{P}\{\mathbb{M}_3 = 3 \mid A_{HH}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = 3\} \cap A_{HH}\}}{\mathbb{P}\{A_{HH}\}} = \frac{p^3}{p^2} = p , \\ \mathbb{P}\{\mathbb{M}_3 = 1 \mid A_{HH}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = 1\} \cap A_{HH}\}}{\mathbb{P}\{A_{HH}\}} = \frac{p^2q}{p^2} = q , \\ \mathbb{P}\{\mathbb{M}_3 = 1 \mid A_{HT}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = 1\} \cap A_{HT}\}}{\mathbb{P}\{A_{HT}\}} = \frac{p^2q}{pq} = p , \\ \mathbb{P}\{\mathbb{M}_3 = -1 \mid A_{HT}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = -1\} \cap A_{HT}\}}{\mathbb{P}\{A_{HT}\}} = \frac{pq^2}{pq} = q , \\ \mathbb{P}\{\mathbb{M}_3 = 1 \mid A_{TH}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = 1\} \cap A_{TH}\}}{\mathbb{P}\{A_{TH}\}} = \frac{p^2q}{pq} = p , \\ \mathbb{P}\{\mathbb{M}_3 = -1 \mid A_{TH}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = -1\} \cap A_{TH}\}}{\mathbb{P}\{A_{TH}\}} = \frac{pq^2}{pq} = q , \\ \mathbb{P}\{\mathbb{M}_3 = -1 \mid A_{TT}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = -1\} \cap A_{TT}\}}{\mathbb{P}\{A_{TT}\}} = \frac{pq^2}{q^2} = p , \\ \mathbb{P}\{\mathbb{M}_3 = -3 \mid A_{TT}\} &= \frac{\mathbb{P}\{\{\mathbb{M}_3 = -1\} \cap A_{TT}\}}{\mathbb{P}\{A_{TT}\}} = \frac{q^3}{q^2} = q . \end{aligned} \quad (34)$$

We then get

$$\begin{aligned}
\mathbb{E}[\mathbb{M}_3 \mid A_{HH}] &= \sum_{y \in \text{Ran}_{A_{HH}}(\mathbb{M}_3)} y \mathbb{P}\{\mathbb{M}_3 = y \mid A_{HH}\} = 3p + 1q = 2 + p - q , \\
\mathbb{E}[\mathbb{M}_3 \mid A_{HT}] &= \sum_{y \in \text{Ran}_{A_{HT}}(\mathbb{M}_3)} y \mathbb{P}\{\mathbb{M}_3 = y \mid A_{HT}\} = 1p + (-1)q = 0 + p - q , \\
\mathbb{E}[\mathbb{M}_3 \mid A_{TH}] &= \sum_{y \in \text{Ran}_{A_{TH}}(\mathbb{M}_3)} y \mathbb{P}\{\mathbb{M}_3 = y \mid A_{TH}\} = 1p + (-1)q = 0 + p - q , \\
\mathbb{E}[\mathbb{M}_3 \mid A_{TT}] &= \sum_{y \in \text{Ran}_{A_{TT}}(\mathbb{M}_3)} y \mathbb{P}\{\mathbb{M}_3 = y \mid A_{TT}\} = (-1)p + (-3)q = -p - 3q = -2 + p - q .
\end{aligned} \tag{35}$$

This means that the random variable $\mathbb{E}[\mathbb{M}_3 \mid \mathcal{A}_2]$ has the following values on the sequences in sample space Ω_3 :

$$\mathbb{E}[\mathbb{M}_3 \mid \mathcal{A}_2](\underline{\omega}) = \begin{cases} 2 + p - q & \underline{\omega} \in A_{HH} \\ p - q & \underline{\omega} \in A_{HT} \\ p - q & \underline{\omega} \in A_{TH} \\ -2 + p - q & \underline{\omega} \in A_{TT} \end{cases} \tag{36}$$

This random variable is the same as $\mathbb{E}[\mathbb{M}_3 \mid \sigma(\mathbb{M}_2)]$.

Exercise 1h (4 points): We repeat the same steps that we had for the calculation of $\mathbb{E}[\mathbb{M}_3 \mid \sigma(\mathbb{M}_2)]$ till the following stage, where we work with $f(\mathbb{M}_3)$ instead of $\mathbb{M}_3$:

$$\begin{aligned}
\mathbb{E}[f(\mathbb{M}_3) \mid B_1] &= \sum_{y \in \text{Ran}_{B_1}(\mathbb{M}_3)} f(y) \mathbb{P}\{\mathbb{M}_3 = y \mid B_1\} = f(3)p + f(1)q , \\
\mathbb{E}[f(\mathbb{M}_3) \mid B_2] &= \sum_{y \in \text{Ran}_{B_2}(\mathbb{M}_3)} f(y) \mathbb{P}\{\mathbb{M}_3 = y \mid B_2\} = f(1)p + f(-1)q , \\
\mathbb{E}[f(\mathbb{M}_3) \mid B_3] &= \sum_{y \in \text{Ran}_{B_3}(\mathbb{M}_3)} f(y) \mathbb{P}\{\mathbb{M}_3 = y \mid B_3\} = f(-1)p + f(-3)q .
\end{aligned} \tag{37}$$

This gives

$$\mathbb{E}[f(\mathbb{M}_3) \mid \sigma(\mathbb{M}_2)](\underline{\omega}) = \begin{cases} f(3)p + f(1)q & \underline{\omega} \in \{\mathbb{M}_2 = 2\} \\ f(1)p + f(-1)q & \underline{\omega} \in \{\mathbb{M}_2 = 0\} \\ f(-1)p + f(-3)q & \underline{\omega} \in \{\mathbb{M}_2 = -2\} \end{cases} \tag{38}$$

Clearly, this is the random variable

$$\mathbb{E}[f(\mathbb{M}_3) \mid \sigma(\mathbb{M}_2)] = f(\mathbb{M}_2 + 1)p + f(\mathbb{M}_2 - 1)q . \tag{39}$$

Thus, the above conditional expectation is a function $g(\mathbb{M}_2)$ where the ordinary function $g(x)$ is written in terms of f as

$$g(x) = f(x + 1)p + f(x - 1)q . \tag{40}$$

The conditional expectation $\mathbb{E}[f(\mathbb{M}_3) \mid \mathcal{A}_2]$ will be the same as earlier, and is equal to $f(x+1)p + f(x-1)q$.

Exercise 1i (2 points): Going by the calculations above, and with no other input, we are able to see a pattern where the value of the conditional expectation $\mathbb{E}[f(\mathbb{M}_N) \mid \mathcal{A}_n]$ is the same as the conditional expectation $\mathbb{E}[f(\mathbb{M}_N) \mid \sigma(\mathbb{M}_n)]$. The basic events in $\sigma(\mathbb{M}_n)$ corresponding to the most-refined partition of Ω_N are either events which are fixed by specifying the first n coin tosses or appropriate unions of those, whereas the basic events in $\mathcal{A}_n$ that form the most-refined partition of Ω_N are simply all those events determined by specifying the first n tosses. It turns out that the conditional expectation on these basic events can be written directly in terms of $f(\mathbb{M}_n + k)$ with k taking values from $N - n$ to $-(N - n)$, with each term weighted by appropriate powers of p and q . This gives the function f_n of $\mathbb{M}_n$.

Exercise 1j (2 points): Choose the random variable $\mathbb{Y}$ to be $\mathbb{Y} = f(\mathbb{M}_N)$, and $\mathcal{G} = \mathcal{A}_n$, $\mathcal{H} = \mathcal{A}_m$ with $m < n$. Since $\mathcal{A}_m \subseteq \mathcal{A}_n$, we use the iterated conditioning property to get

$$\mathbb{E}\left[\mathbb{E}[f(\mathbb{M}_N) \mid \mathcal{A}_n] \mid \mathcal{A}_m\right] = \mathbb{E}[f(\mathbb{M}_N) \mid \mathcal{A}_m] , \quad (41)$$

which is the same as

$$\mathbb{E}[f_n(\mathbb{M}_n) \mid \mathcal{A}_m] = f_m(\mathbb{M}_m) . \quad (42)$$

This is the martingale condition for the stochastic process $f_n(\mathbb{M}_n)$, $0 \leq n \leq N$.

Some material on conditional expectations and Markov processes

A discrete stochastic process $\mathbb{X}_n$ is a Markov process if, for any given k between 1 and n , and any choice of time instants $n_1, n_2, \dots, n_k$ between 0 and n , the following conditional probabilities should be equal:

$$\mathbb{P}\{\mathbb{X}_n = a \mid \mathbb{X}_{n_k} = a_k, \dots, \mathbb{X}_{n_2} = a_2, \mathbb{X}_{n_1} = a_1\} = \mathbb{P}\{\mathbb{X}_n = a \mid \mathbb{X}_{n_k} = a_k\} . \quad (43)$$

In other words, the conditional probability that $\mathbb{X}_n = a$ depends only on the previously measured value $\mathbb{X}_{n_k}$ at time n_k and none of the values at the other previous time instants.

The above can be phrased in terms of σ -algebras as follows. Fix a time instant n , and consider *all* events of the form

$$\{\mathbb{X}_n = a, \mathbb{X}_{n_k} = a_k, \dots, \mathbb{X}_{n_2} = a_2, \mathbb{X}_{n_1} = a_1\} , \quad (44)$$

for all choices of the number of instants k , the time instants $n > n_k > n_{k-1} > \dots > n_2 > n_1 > 0$, and all choices of the locations $a_k, a_{k-1}, \dots, a_2, a_1$ at the above time instants. Clearly, this is a huge set of events, and they are all determined by the information of the path of the random process up to time n . These events all belong to the σ -algebra $\mathcal{A}_n$ generated by

subsets of the form $\{\mathbb{X}_n = a_n, \mathbb{X}_{n-1} = a_{n-1}, \dots, \mathbb{X}_2 = a_2, \mathbb{X}_1 = a_1\}$ for all possible allowed values $a_n, a_{n-1}, \dots, a_2, a_1$.

We saw an example of this in Assignment 2 where we considered the σ -algebra $\mathcal{A}_n$ generated by the paths of the random walk up to time n . We showed that $\{\mathcal{A}_n\}_{0 \leq n \leq N}$ form a filtration of σ -algebras. The random walk process $\mathbb{M}_n$ is adapted to this filtration since the σ -algebra generated by $\mathbb{M}_n$ is a σ subalgebra of $\mathcal{A}_n$. Another example is the stock price process $\mathbb{S}_n = \mathbb{S}_0 u^{\mathbb{M}_n}$ which is also adapted to the same filtration $\{\mathcal{A}_n\}$ since $\mathbb{S}_n$ is simply the exponential of $\mathbb{M}_n$.

Recall that the σ -algebra $\sigma(\mathbb{X})$ generated by a random variable $\mathbb{X}$ is obtained by looking at events in the sample space of the form $\{\omega \in \Omega \mid \mathbb{X}(\omega) \in A\}$ for all possible Borel subsets A of $\mathbb{R}$, and then taking countable unions, intersections and complements. The set $\{\omega \in \Omega \mid \mathbb{X}(\omega) \in A\}$ is usually shortened to $\{\mathbb{X} \in A\}$ or $\mathbb{X}^{-1}(A)$. Technically, $\sigma(\mathbb{X})$ is the smallest σ -algebra that contains sets of the form $\{\mathbb{X} \in A\}$. In the current situation where we work with discrete random variables which take only a finite number of values, $\sigma(\mathbb{X})$ can be generated by sets of the form $\{\mathbb{X} = b\}$ where b runs through all the finite values that $\mathbb{X}$ can take.

Next, if a random variable $\mathbb{Y}$ is $\sigma(\mathbb{X})$ -measurable, it can be shown that the random variable $\mathbb{Y}$ can be written as $f(\mathbb{X})$ for some function f . Saying $\mathbb{Y}$ is $\sigma(\mathbb{X})$ -measurable means that $\sigma(\mathbb{Y})$ is a σ subalgebra of $\sigma(\mathbb{X})$. That is, every set of the form $\{\mathbb{Y} \in A\}$ is contained in $\sigma(\mathbb{X})$. More descriptively, every set of the form $\{\mathbb{Y} \in A\}$ is of the form $\{\mathbb{X} \in B\}$. In particular, the event $\{\mathbb{Y} = a\}$ corresponds to an event in $\sigma(\mathbb{X})$ of the form $\{\mathbb{X} \in B_a\}$, where the subscript a on B_a simply conveys that it arises in considering subsets of the sample space of the form $\{\mathbb{Y} = a\}$. It is easy to see that, for two values $a \neq b$, the Borel subsets B_a and B_b must be disjoint, that is $B_a \cap B_b = \emptyset$. Otherwise, there will be at least one element ω such that $\mathbb{Y}(\omega) = a$ as well as $\mathbb{Y}(\omega) = b$ which is not possible for any function on the sample space.

The above discussion actually allows to express $\mathbb{Y}$ as a function of $\mathbb{X}$. This function is constructed as follows. Let $\mathbb{I}_{B_a}(x)$ be the indicator function for the Borel subset B_a , which means $\mathbb{I}_{B_a}(x) = 1$ when $x \in B_a$ and is zero otherwise. Consider the function

$$f(x) = \sum_{a \in \text{Ran}(\mathbb{Y})} a \mathbb{I}_{B_a}(x) . \quad (45)$$

It can be verified that $f(\mathbb{X})$ is the same random variable as $\mathbb{Y}$. That is, $\mathbb{Y}(\omega) = f(\mathbb{X}(\omega))$ for every ω in the sample space Ω . Thus, the upshot is that if $\mathbb{Y}$ is $\sigma(\mathbb{X})$ -measurable, then it can be written as a function of $\mathbb{X}$. This can be extended to the situation where $\mathbb{Y}$ is measurable with respect to $\sigma(\mathbb{X}_1, \mathbb{X}_2, \dots, \mathbb{X}_n)$, the σ -algebra generated by a collection of random variables $\mathbb{X}_1, \mathbb{X}_2, \dots, \mathbb{X}_m$. In this case, $\mathbb{Y}$ can be written as some function of the n random variables $\mathbb{X}_1, \dots, \mathbb{X}_m$.

We can apply this to the case of the conditional expectation $\mathbb{E}[Z \mid \sigma(\mathbb{X}_m)]$ which is a $\sigma(\mathbb{X}_m)$ -measurable random variable. The previous discussion implies that it is simply a function $f(\mathbb{X}_m)$ of the random variable $\mathbb{X}_m$. Recall the procedure to compute the values of the random variable $\mathbb{E}[Z \mid \sigma(\mathbb{X}_m)]$: First, we obtain the most-refined partition of the sample space, where the partition is made up of events in $\sigma(\mathbb{X}_m)$. A convenient choice is the collection of events $B_b = \{\mathbb{X}_m = b\}$ for all possible allowed values b of $\mathbb{X}_m$. Next, we compute the conditional expectation values $\mathbb{E}[Z \mid B_b]$ which are given as

$$\mathbb{E}[Z \mid B_b] = \sum_{y \in \text{Ran}_{B_b}(\mathbb{Z})} y \mathbb{P}\{Z = y \mid B_b\} . \quad (46)$$

Suppose the possible values of $\mathbb{X}_m$ are $\{b_1, b_2, \dots\}$. The value of the random variable $\mathbb{E}[Z \mid \sigma(\mathbb{X}_m)]$ on an element ω in Ω is then

$$\mathbb{E}[Z \mid \sigma(\mathbb{X}_m)](\omega) = \begin{cases} \mathbb{E}[Z \mid B_{b_1}] & \omega \in B_{b_1} \\ \mathbb{E}[Z \mid B_{b_2}] & \omega \in B_{b_2} \\ \vdots & \vdots \end{cases} \quad (47)$$

This is simply the function of $\mathbb{X}_m$ obtained by calculating the expectation values

$$\mathbb{E}[Z \mid \mathbb{X}_m] \quad \text{for every allowed value of } \mathbb{X}_m. \quad (48)$$

This has the same form as the general equation (45) in terms of indicator functions, but we will not write it that way since it requires a lot of notation which will not be useful for the rest of the exercise.

The upshot of the above discussion is that

$$\mathbb{E}[Z \mid \sigma(\mathbb{X}_m)] = \mathbb{E}[Z \mid \mathbb{X}_m] . \quad (49)$$

Similarly, since the σ -algebra $\mathcal{A}_m$ is generated by $\mathbb{X}_m, \mathbb{X}_{m-1}, \dots, \mathbb{X}_1$, the conditional expectation $\mathbb{E}[Z \mid \mathcal{A}_m]$ has the explicit description

$$\mathbb{E}[Z \mid \mathcal{A}_m] = \mathbb{E}[Z \mid \mathbb{X}_m, \mathbb{X}_{m-1}, \dots, \mathbb{X}_1] , \quad (50)$$

where the right hand side is understood to be a function of the values of the random variables $\mathbb{X}_m, \mathbb{X}_{m-1}, \dots, \mathbb{X}_1$.

Let us recall what it means for a process $\mathbb{X}_n$ to be Markov. The stochastic process $\mathbb{X}_n$, $0 \leq n \leq N$ is Markov if the following relation between conditional probabilities holds:

$$\mathbb{P}\{\mathbb{X}_n = a \mid \mathbb{X}_{n_m} = a_m, \dots, \mathbb{X}_{n_1} = a_1\} = \mathbb{P}\{\mathbb{X}_n = a \mid \mathbb{X}_{n_m} = a_m\} , \quad (51)$$

for every choice of $m \leq n$ and every choice of the m time instants $n_m, n_{m-1}, \dots, n_1$ below n , and every choice of allowed values $a_m, a_{m-1}, \dots, a_1$. This allows us to simplify the

conditional expectation values further. Choosing the random variable $\mathbb{Y}$ to be $\mathbb{X}_n$, we have the conditional expectation value

$$\begin{aligned} \mathbb{E}[\mathbb{X}_n \mid \mathbb{X}_m = a_m, \mathbb{X}_{m-1} = a_{m-1}, \dots, \mathbb{X}_1 = a_1] \\ = \sum_{y \in \text{Ran}(\mathbb{X}_n)} y \mathbb{P}\{\mathbb{X}_n = y \mid \mathbb{X}_m = a_m, \mathbb{X}_{m-1} = a_{m-1}, \dots, \mathbb{X}_1 = a_1\} , \end{aligned} \quad (52)$$

where $\text{Ran}(\mathbb{X}_n)$ is the range of values of $\mathbb{X}_n$.¹ Using the Markov property, the right hand side of the above equation becomes

$$\begin{aligned} \sum_{y \in \text{Ran}(\mathbb{X}_n)} y \mathbb{P}\{\mathbb{X}_n = y \mid \mathbb{X}_m = a_m, \mathbb{X}_{m-1} = a_{m-1}, \dots, \mathbb{X}_1 = a_1\} \\ = \sum_{y \in \text{Ran}(\mathbb{X}_n)} y \mathbb{P}\{\mathbb{X}_n = y \mid \mathbb{X}_m = a_m\} . \end{aligned} \quad (53)$$

We have proved, for every allowed set of values $a_m, a_{m-1}, \dots, a_1$, that

$$\mathbb{E}[\mathbb{X}_n \mid \mathbb{X}_m = a_m, \mathbb{X}_{m-1} = a_{m-1}, \dots, \mathbb{X}_1 = a_1] = \mathbb{E}[\mathbb{X}_n \mid \mathbb{X}_m = a_m] . \quad (54)$$

That is,

$$\mathbb{E}[\mathbb{X}_n \mid \mathcal{A}_m] = \mathbb{E}[\mathbb{X}_n \mid \sigma(\mathbb{X}_m)] = \mathbb{E}[\mathbb{X}_n \mid \mathbb{X}_m] . \quad (55)$$

The last equality in (55) is very useful for explicit calculations. It is not hard to extend our arguments to the case where we have a function $f(\mathbb{X}_n)$ instead of $\mathbb{X}_n$ itself:

$$\mathbb{E}[f(\mathbb{X}_n) \mid \mathcal{A}_m] = \mathbb{E}[f(\mathbb{X}_n) \mid \sigma(\mathbb{X}_m)] = \mathbb{E}[f(\mathbb{X}_n) \mid \mathbb{X}_m] = g(\mathbb{X}_m) . \quad (56)$$

¹Going by the formula given in the assignment sheet, the sum should be over the restricted range $\text{Ran}_{\vec{a}}(\mathbb{X}_n)$ obtained by restricting the domain of $\mathbb{X}_n$ to the subset of the sample space dictated by the conditions $\mathbb{X}_m = a_m, \mathbb{X}_{m-1} = a_{m-1}, \dots, \mathbb{X}_1 = a_1$ (the vector $\vec{a}$ is shorthand for the values $a_m, a_{m-1}, \dots, a_1$). In fact, this is not strictly necessary since the conditional probabilities for y outside this range will be zero. Hence, even if we sum over the full range of $\mathbb{X}_n$, only those values will be picked out whose conditional probabilities are not zero.